

The Fixed-Coordinate Gauge Obstruction

Identical Relative Tails with Arbitrarily Bad Loewner Transfer

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Abstract

We prove that the gauge in RH-120 cannot be dropped. For every $0 < \varepsilon \leq 1$ there are positive four-dimensional pairs (G, D) and (G', D') with identical relative Rayleigh constants and identical generalized spectra, related by a coordinate swap, while the best fixed-coordinate RH-120 factor is exactly $1/\varepsilon$. The swapping gauge has factor one. Thus spectral agreement of the two relative pairs does not control a common-coordinate Loewner comparison. A 512-point audit from 10^{-1} to 10^{-14} has zero failures and log-log slope -1 . This is a sharp negative theorem, not a failure of the gauge-covariant route.

1 Best constants without a gauge

For $G, G', D, D' \succ 0$, the largest a and smallest b satisfying

$$G' \succeq aG, \quad D' \preceq bD$$

are

$$a = \lambda_{\min}(G^{-1/2}G'G^{-1/2}), \quad b = \lambda_{\max}(D^{-1/2}D'D^{-1/2}).$$

RH-120 then gives $\gamma' \leq \sqrt{b/a} \gamma$. These constants are optimal by the variational characterization of ordered eigenvalues [Bhatia, 2007, Horn and Johnson, 1991].

2 Sharp swapped-axis obstruction

Theorem 1 (Unbounded fixed-coordinate loss). *Fix $\gamma > 0$ and $0 < \varepsilon \leq 1$. Let*

$$G_\varepsilon = \text{diag}(\varepsilon, 1, 2, 3), \quad D_\varepsilon = \gamma^2 G_\varepsilon,$$

and let P swap the first two coordinates. Put $G'_\varepsilon = P^ G_\varepsilon P$ and $D'_\varepsilon = P^* D_\varepsilon P$. Then*

$$\gamma(G_\varepsilon, D_\varepsilon) = \gamma(G'_\varepsilon, D'_\varepsilon) = \gamma,$$

but the best identity-gauge constants are $a = \varepsilon$ and $b = \varepsilon^{-1}$. Hence the fixed-coordinate transfer factor is $\sqrt{b/a} = \varepsilon^{-1}$.

Proof. Both relative matrices equal $\gamma^2 I$, so the Rayleigh constants agree. The generalized eigenvalues of (G', G) contain ε and ε^{-1} , giving $a = \varepsilon$. Since $D' = \gamma^2 G'$ and $D = \gamma^2 G$, the corresponding upper factor is $b = \varepsilon^{-1}$. $\square$

Corollary 2 (Gauge recovery). *Using $S = P$ gives $G' = S^* G S$ and $D' = S^* D S$, so RH-120 transfers with $a = b = 1$ and no loss.*

The example is deliberately stronger than a mismatch of scalar norms. The entire generalized spectrum is unchanged. What fails is the unproved identification of two anisotropic frame coordinates.

3 Audit and route consequence

We evaluate the family at 512 logarithmically spaced anisotropies. The computed identity-gauge factor agrees with $1/\varepsilon$, source and target gammas agree, and the swapping gauge has factor one in every case. The fitted log-log slope is -1 to machine precision.

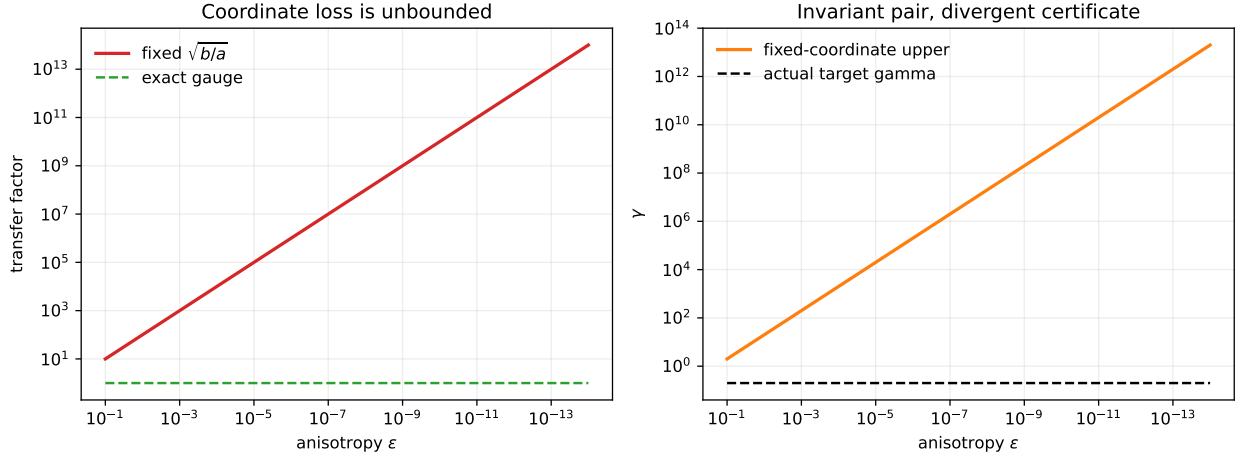


Figure 1: The physical relative pair is invariant, while a forced common coordinate system creates an arbitrarily bad certificate.

RH-122 closes the no-gauge branch: a future cross-level proof must either construct a gauge or prove an additional common-coordinate coherence law. The negative result leaves the gauge-covariant route intact and motivates the defect-stable recurrence of RH-123. No uniform physical gauge, Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis claim is made.

References

- Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.
- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.